

Agri-Sahayak Research Papers — Dataset Link Directory

Compiled from the 66-paper reference collection (agri_research_papers_updated.xlsx). Each entry lists any dataset repository link, source, or data-availability note found directly inside that paper's PDF text.

Paper #1

Incorporating soil information with machine learning for crop recommendation to improve agricultural output

Module: 4.1 Crop Recommendation

- Crop recommendation dataset (Indian Chamber of Food & Agriculture, via Kaggle)
<https://www.kaggle.com/> (search: Indian Chamber of Food and Agriculture crop dataset)

Paper #2

Machine learning based recommendation using NPK, soil pH and three climatic variables

Module: 4.1 Crop Recommendation

- Crop dataset (Kaggle, general reference)
<https://www.kaggle.com/>

Paper #3

Enhancing Agricultural Productivity: A ML Approach to Crop Recommendations

Module: 4.1 Crop Recommendation

- Dataset generated during study
Not public — available from corresponding author upon reasonable request

Paper #5

Crop Recommendation with ML: Leveraging Environmental and Economic Factors for Optimal Crop Selection

Module: 4.1 Crop Recommendation

- Crop yield recommendation dataset (Kaggle)
<https://www.kaggle.com/datasets/vihith12/crop-yield-recommendation-dataset>

Paper #6

Smart Profit-Aware Crop Advisory System: Kisan AI (RF + Prophet + Market Price)

Module: 4.1 Profit-Aware

- Crop Recommendation Dataset (Kaggle)
<https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset/data>
- Fertilizer Prediction Dataset (Kaggle)
<https://www.kaggle.com/datasets/mohitsingh1804/fertilizer-prediction-dataset>
- PlantVillage (TensorFlow Datasets catalog)
https://www.tensorflow.org/datasets/catalog/plant_village
- Agmarknet Government Market Price Portal (India)
<https://agmarknet.gov.in/>

Paper #8

Optimizing agricultural yield: a predictive model for profitable crop harvesting based on market dynamics

Module: 4.1 Profit-Aware

- Dataset availability

Data availability statement present — check paper for details (not a public link)

Paper #10

Enhancing global agricultural monitoring system for climate-smart agriculture (NDVI, multispectral)

Module: 4.2 Weather Intelligence

- ■ NOTE: This PDF file is a duplicate of paper #9's PDF (upload mismatch)

Re-upload correct PDF for #10 if needed.

Paper #11

ASAP: A new global early warning system to detect anomaly hot spots of agricultural production

Module: 4.2 Weather Intelligence

- Climate/agriculture dataset

Data will be made available on request

Paper #14

Plant Leaf Disease Detection Using Deep Learning: A Multi-Dataset Approach

Module: 4.3 Disease Detection

- Plant pest/disease datasets (strawberry, rice, etc.)

Referenced as publicly available on GitHub — see paper's citation list for individual links; also 'available upon request'

Paper #15

A systematic review of deep learning techniques for plant diseases (2020–2024)

Module: 4.3 Disease Detection

- PlantDoc Dataset

<https://github.com/pratikkayal/PlantDoc-Dataset>

- Review's own extracted data

<https://zenodo.org/records/14051480>

Paper #16

A Review on Detection of Plant Disease Using ML and Deep Learning Approaches

Module: 4.3 Disease Detection

- PlantVillage (Hughes & Salathé 2015)

<https://www.kaggle.com/abdallahalidev/plantvillage-dataset>

- AI Challenger 2018

<https://pan.baidu.com/s/1TH9qL7Wded2Qiz03wHTDLw#list/path=%2F>

- Plant Diseases 2018

<https://www.kaggle.com/saroz014/plant-diseases>

- Cassava Leaf Disease Classification 2019

Private dataset — not public

- Citrus Leaves 2020

<https://www.kaggle.com/dtrilsbeek/citrus-leaves-prepared>

- PlantDoc 2021
<https://www.kaggle.com/datasets/abdulhasibuddin/plant-doc-dataset>
- PlantPathology 2020 (Kaggle Competition)
<https://www.kaggle.com/c/plant-pathology-2020-fgvc7>
- Grapevine (Koklu et al. 2022)
<https://www.muratkoklu.com/datasets/>

Paper #17

Leveraging deep learning for plant disease identification: bibliometric analysis 2018–2024

Module: 4.3 Disease Detection

- N/A
- Data Availability Statement: No new data were created in this study

Paper #18

IoT Sensing for Advanced Irrigation Management: A Systematic Review of Trends, Challenges, and Future Prospects

Module: 4.4 Irrigation Advisory

- Review's compiled dataset
<https://github.com/Enowtakang/Review2024>

Paper #20

Recent advances in IoT-driven crop monitoring and precision irrigation

Module: 4.4 Irrigation Advisory

- IoT/agri sensor dataset
- Data will be made available on request

Paper #24

Enhancing agricultural commodity price forecasting with deep learning (LSTM, GRU)

Module: 4.5 Market Price

- Discusses data availability as a general limitation across reviewed studies
- No single dataset — SLR paper, see individual cited studies

Paper #26

Various optimized machine learning techniques to predict agricultural commodity prices

Module: 4.5 Market Price

- AgriPriceBD — Bangladeshi commodity price dataset (Mendeley Data)
<https://data.mendeley.com/datasets/bkmxnrn3hn>
- Codebase / experiment infrastructure (GitHub)
<https://github.com/TashreefMuhammad/Bangladesh-Agri-Price-Forecast>

Paper #27

Predicting Agricultural Commodities Prices with Machine Learning: A Review of Current Research

Module: 4.5 Market Price

- Dataset generated/analyzed during study
- Available from corresponding author on reasonable request

Paper #28

Democratising Agricultural Commodity Price Forecasting: The AGRICAF Approach

Module: 4.5 Market Price

- N/A

General discussion of data availability as a challenge, no specific dataset

Paper #29

Smart Profit-Aware Crop Advisory System: Kisan AI (Integrated Dashboard)

Module: 4.6 Dashboard

- Global commodity price data

World Bank (Pink Sheet), FAO, USDA/FAS — see paper references, no single direct link

Paper #30

ASAP: Global Early Warning System with Disease-Risk Visualization

Module: 4.6 Dashboard

- ■ NOTE: This PDF file is a duplicate of paper #6's PDF (upload mismatch)

See paper #6 entry — original title for #30 in your list is 'ASAP: Global Early Warning System', but file content is Kisan AI paper. Re-upload correct PDF for #30 if needed.

Paper #31

Empowering Rural Livestock Health: AI-Powered Early Detection of Cattle Diseases

Module: 4.7 Livestock Advisory

- ■ NOTE: This PDF file is a duplicate of paper #12's PDF (upload mismatch)

Re-upload correct PDF for #31 if needed.

Paper #32

AI-DISA: Artificial Intelligence-Based Disease Identification System for Livestock (CNN, YOLO, ResNet)

Module: 4.7 Livestock Advisory

- Lumpy Skin Images Dataset (Kaggle)

<https://www.kaggle.com/datasets/warcoder/lumpy-skin-images-dataset>

- Cow Lumpy Disease Dataset (Kaggle)

<https://www.kaggle.com/datasets/shivamagarwal29/cow-lumpy-disease-dataset>

Paper #34

AI-powered visual E-monitoring system for cattle health and wealth

Module: 4.7 Livestock Advisory

- Lumpy Skin Images Dataset (Mendeley Data)

<https://data.mendeley.com/datasets/w36hpf86j2>

Paper #36

Design of agricultural product traceability system based on blockchain and RFID

Module: 4.8 Traceability

- FAO Food and Agriculture Data — Crops & Livestock

<https://www.fao.org/faostat/en/>

Paper #38

Digital Transformation of Food Supply Chain Using Blockchain: Systematic Literature Review

Module: 4.8 Traceability

- No explicit dataset link found

Paper #42

Customised agricultural advice at scale: How digital extension helps Indian farmers

Module: 4.9 Accessibility

- No explicit dataset link found

Paper #45

A Machine Learning Approach for Crop Yield and Disease Prediction Integrating Soil Nutrition and Weather Factors (SRDI + SARIMAX + Bangladesh AEZ — The CUET Reference Paper)

Module: 4.1b Soil Health Prediction

- SRDI soil data / Bangladesh AEZ; World Bank open data reference
<https://blogs.worldbank.org/en/opendata/mind-gap-enabling-data-> (context reference, not the dataset itself)

Paper #46

An IoT-enabled AI system for real-time crop prediction using soil and weather data in precision agriculture (SRDI + Bangladesh AEZ + IoT)

Module: 4.1b Soil Health Prediction

- Bangladesh Meteorological Department (BMD) — weather data source
<https://bmd.gov.bd/>

Paper #47

The IoT and AI in Agriculture: The Time Is Now — A Systematic Review of Smart Sensing Technologies (2020–2024)

Module: 4.1b Soil Health Prediction

- Implementation code + private crop prediction dataset (GitHub)
<https://github.com/SAIFULLAH-SHARAFAT/An-IoT-Enabled-AI-System-for-Real-Time-Crop-Prediction-Using-Soil-and-Weather-Data>

Paper #50

Legume-based rotation benefits crop productivity and agricultural sustainability in the North China Plain

Module: 4.11 Crop Rotation Engine

- No explicit dataset link found

Paper #52

Flood-tolerant rice improves climate resilience, profitability, and household consumption in Bangladesh (BRRI dhan51, dhan52, BINA Dhan 11, Dhan 12)

Module: 4.12 AEZ Localization

- Data within paper's Supporting Information files
Not an external link — embedded in the paper's supplementary files

Paper #56

Piloting a Weather-Index-Based Crop Insurance System in Bangladesh: Challenges of Financial Instruments for Tackling Climate Risks (WIBCI + bKash)

Module: 4.10 Financial Inclusion

- No explicit dataset link found

Paper #57

Leveraging index insurance to protect farmers from weather-based risk

Module: 4.10 Financial Inclusion

- WIBCI pilot project data
Available from the first author upon reasonable request (privacy-restricted)

Paper #60

Aggregation models and small farm commercialization: A scoping review of global literature

Module: 4.5b Cooperative Selling

- No explicit dataset link found

Paper #62

AI for Data-Driven Decision-Making in Smart Agriculture: From Field to Farm Management

Module: Global Research

- CGIAR-affiliated review — no single dataset (literature review)

Paper #64

On-farm performance of genetically-improved rohu (*Labeo rohita*) in Bangladesh

Module: F1 — Major Carps (Rohu/Catla/Mrigal)

- Fish farm operational data
Available upon request from corresponding author

Paper #66

Profit efficiency of Pangas (*Pangasianodon hypophthalmus*) pond fish farming in Bangladesh

Module: F2 — Exotic Carps (Pangasius/Tilapia/Silver/Grass/Common Carp)

- No explicit dataset link found

Paper #69

Comparative juvenile performance assessment of genetically improved Nile tilapia (GIFT) under commercial conditions in Bangladesh

Module: F2 — Exotic Carps (Pangasius/Tilapia/Silver/Grass/Common Carp)

- Fish stock/GIFT matching dataset
Available from authors upon request

Paper #78

Hilsa fishery management in Bangladesh — HFMAP & ECOFISH science-based adaptive co-management

Module: F5 — Hilsa / Ilish Fishery & Conservation

- Data availability statement present
- [Check paper directly — no link extracted](#)

Paper #79

Assessing WSSV and AHPND concurrent with *Vibrio* spp. in *Penaeus monodon* farms in southwestern Bangladesh

Module: F6 — Shrimp Aquaculture (Bagda/WSSV/Gher System)

- No explicit dataset link found

Paper #83

Current Status and Prospects of Farming the Giant River Prawn (*Macrobrachium rosenbergii*) in Bangladesh

Module: F7 — Freshwater Prawn (Golda / *Macrobrachium rosenbergii*)

- No explicit dataset link found

Paper #86

Linking human nutrition and fisheries: incorporating micronutrient-dense SIS in carp polyculture in Bangladesh

Module: F8 — Small Indigenous Species (SIS) & Nutrition

- No explicit dataset link found

Paper #92

Stock assessment of mud crab (*Scylla olivacea*) from Sundarbans mangrove forest, Bangladesh

Module: F9 — Mud Crab (*Scylla serrata* / Kakra)

- Data used in manuscript
- [Available on request](#)

Quick Reference — All Direct Download Links

Every clickable dataset/repository URL extracted from the papers, in one place.

Paper #	Dataset	Link
1	Crop recommendation dataset (Indian Chamber of Food & Agriculture, via Kaggle)	https://www.kaggle.com/ (search: Indian Chamber of Food and Agriculture crop dataset)
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5	Crop yield recommendation dataset (Kaggle)	https://www.kaggle.com/datasets/vihith12/crop-yield-recommendation-dataset
6	Crop Recommendation Dataset (Kaggle)	https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset/data
6	Fertilizer Prediction Dataset (Kaggle)	https://www.kaggle.com/datasets/mohitsingh1804/fertilizer-prediction-dataset
6	PlantVillage (TensorFlow Datasets catalog)	https://www.tensorflow.org/datasets/catalog/plant_village
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32	Lumpy Skin Images Dataset (Kaggle)	https://www.kaggle.com/datasets/warcoder/lumpy-skin-images-dataset
32	Cow Lumpy Disease Dataset (Kaggle)	https://www.kaggle.com/datasets/shivamagarwal29/cow-lumpy-disease-dataset
34	Lumpy Skin Images Dataset (Mendeley Data)	https://data.mendeley.com/datasets/w36hpf86j2
36	FAO Food and Agriculture Data — Crops & Livestock	https://www.fao.org/faostat/en/

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47	Implementation code + private crop prediction dataset (GitHub)	https://github.com/SAIFULLAH-SHARAFAT/An-IoT-Enabled-AI-System-for-Real-Time-Crop-Prediction-Using-Soil-and-Weather-Data